Producers and consumers

The modern concept of a food chain explains that some organisms produce their own food. These are known as producers, or autotrophs. Plants and most algae fall into this category, normally using the energy of sunlight to convert water and carbon dioxide into glucose, at the same time as releasing oxygen. This process, photosynthesis, is the first step towards creating food. In places where there is no sunlight, organisms producing their own food are called chemotrophs. Those in the deep ocean, for example, get the energy they need from chemicals released by hydrothermal vents.

Animals that eat producers and creatures that eat other animals are called consumers, or heterotrophs. There may be two, three, or more levels of these in any particular part of the food chain, but there is always a producer at the bottom, and all levels above it are consumers. Animals that only eat plants are herbivores, or primary consumers, and they include cattle, rabbits, butterflies, and elephants. Those that eat only other animals are carnivores, or secondary consumers; these include thrushes, dragonflies, and hedgehogs. In turn, secondary consumers may be eaten by larger predators, or tertiary consumers, such as foxes, small cats, and birds of prey. The animals at the top of their food chain are apex predators. They include consumers such as tigers, killer whales, and golden eagles that are not preyed upon by other animals.

The food chain does not break when plants and animals die. Detritus feeders (detritivores) prey on the remains, recycling nutrients and energy for the next generation of producers to use.